

Energy band gap of semiconductor

①## Aim

To determine the energy band gap of a semiconductor p-n junction diode

②## Apparatus

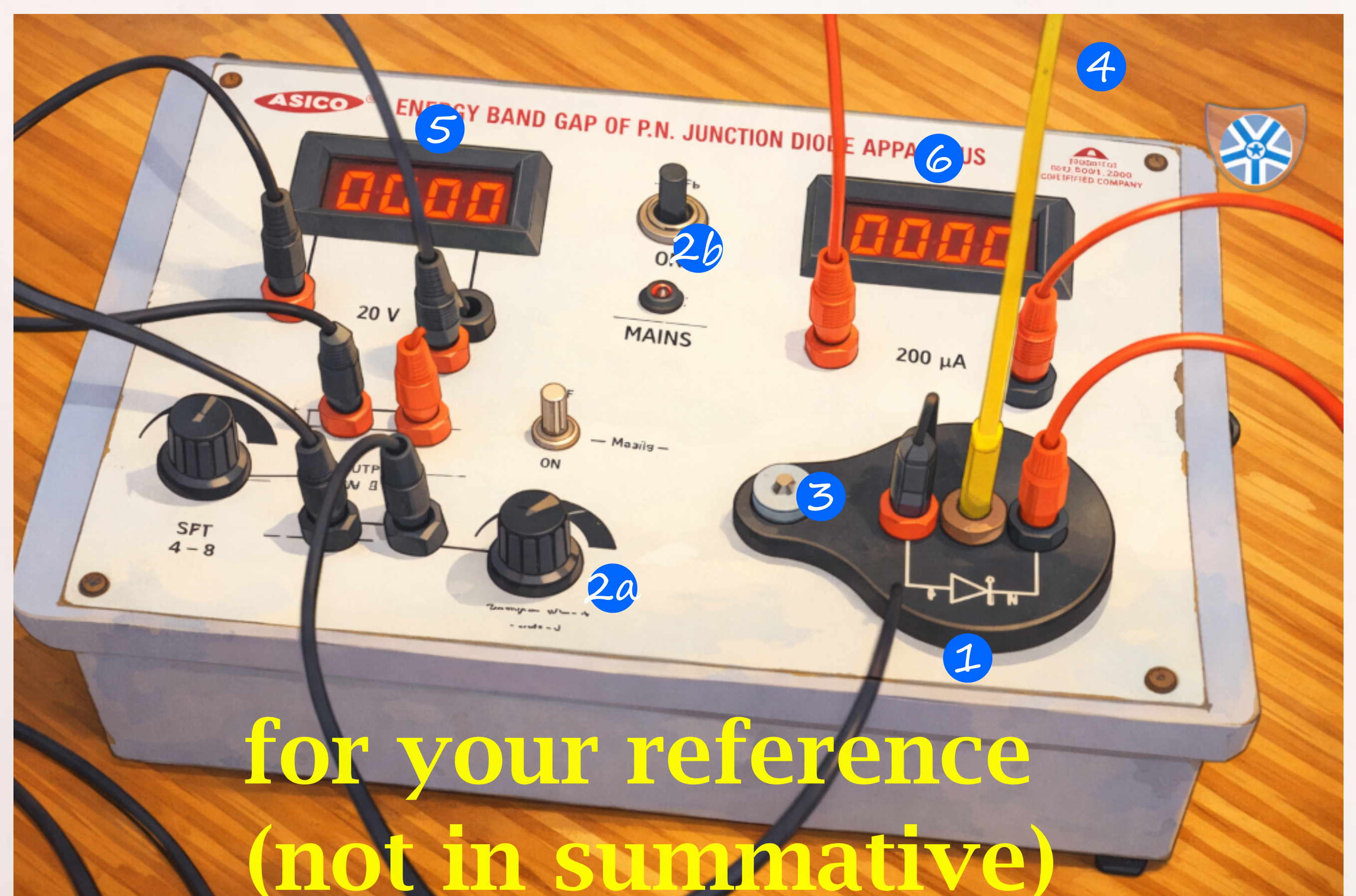
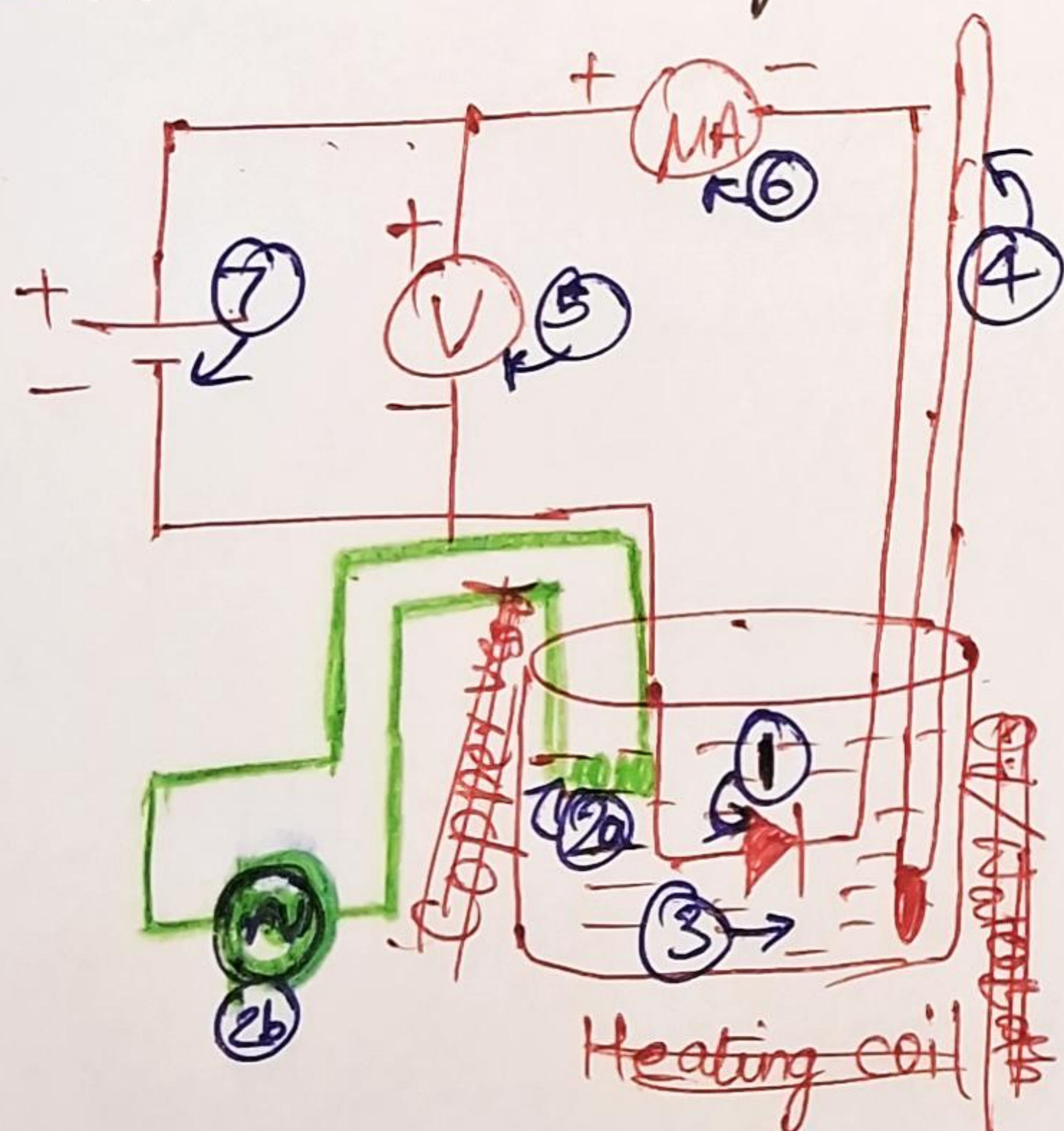
1. Germanium diode
2. Regulated DC powersupply
3. Voltmeter
4. Ammeter
5. Thermometer
6. Temperature controller
7. Copper vessel
8. Heater
9. Connecting wires

③## Formula

$$E_g = \frac{2 \times 2.302 \times K_B \times \text{slope} \times 1000}{1.6 \times 10^{-19}}$$

Quantity	Symbol	Unit
Energy bandgap E_g	E_g	eV
Boltzmann constant	K_B	$1.38 \times 10^{-23} \text{ J/K}$
Slope of $\log_{10} I_v$ vs $\frac{1000}{T}$	slope	K

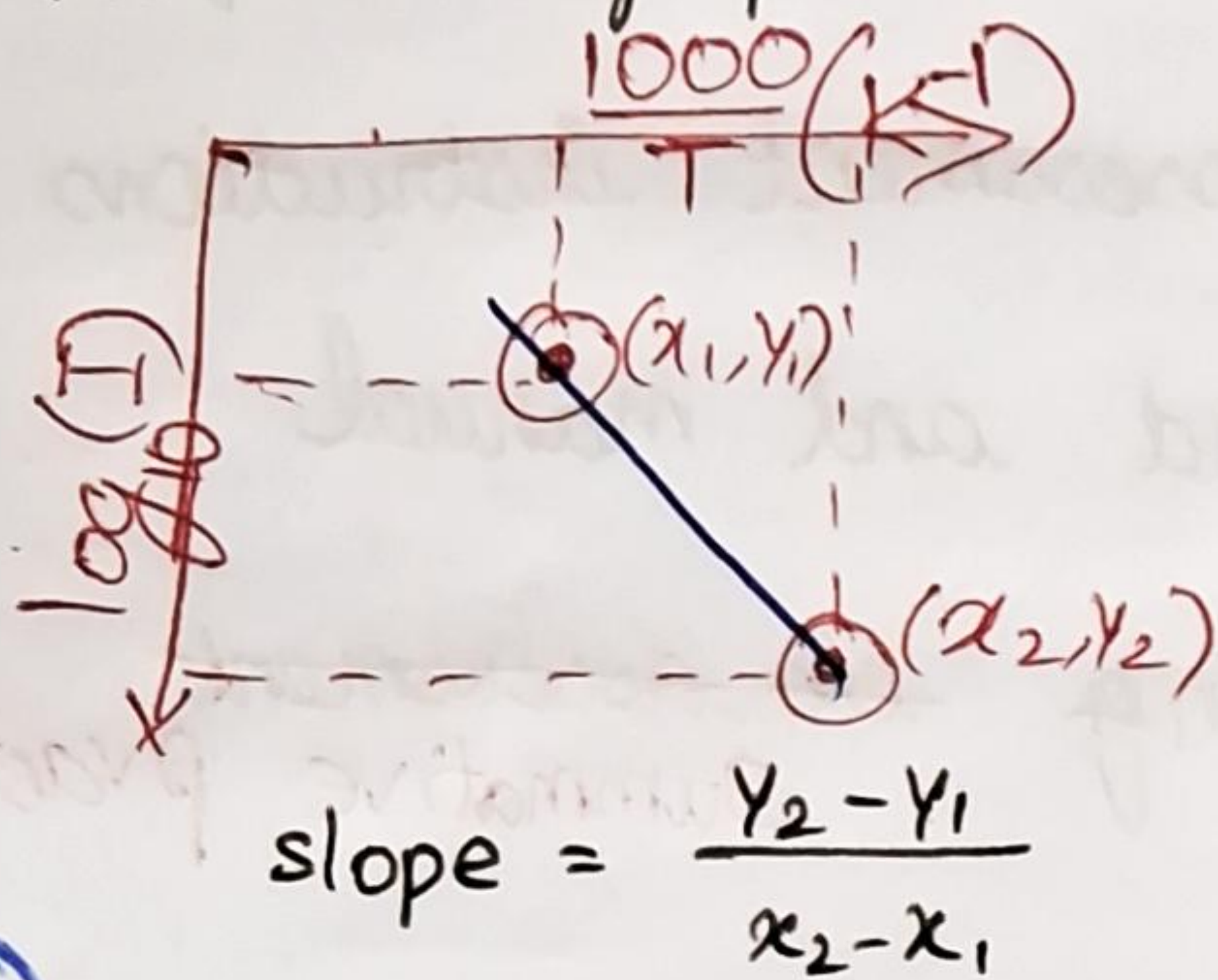
④## Circuit diagram



for your reference
(not in summative)

- ① Germanium pn junction diode
- ② a Heating coil
- ③ Oil bath
- ④ Thermometer
- ⑤ Voltmeter
- ⑥ Ammeter
- ⑦ Battery

5. Model graph



6. Tabular form

S.No	Temperature (T)		$\frac{1000}{T}$ (K ⁻¹)	Current I (μ A)	$\log I$
	T (°C)	T+273 (K)			
1					
2					
...					

7. Procedure

1. Place the pn junction diode inside a copper vessel.
2. Copper vessel has oil. Heat the oil with heater connected to AC main.
3. Measure the temperature of oil bath with thermometer.
4. Connect the pn junction to DC power supply under reverse bias voltage of 1.5 V and current readings.
5. Heat the diode by and take temperature readings every 5°C upto 70°C. This is called heating curve. Switch off AC mains and let the diode cool. Take the temperature and current every 5°C decrease of temperature. This is called cooling curve.

8. Precautions

1. Note the temperature and current readings simultaneously.
2. Connect the diode in reverse bias condition.

9. Calculations

Result

10. Applications

- Pn junction diode under reverse bias is a good temperature sensor.

Viva (External)

- What is energy band gap?
- What is Fermi level?
- If temperature increases, what happens to energy band gap?